

Assignment - 2

Function: —

(*) Draw the graph and test whether they are function or not (using vertical line test):

(i) $y = -x^2 + 2$ (ii) $y = -|x| - 3$ (iii) $y = -|x+2| + 2$

(iv) $y = \sqrt{x-2} + 2$ (v) $y = (x+2)^2 + 1$ (vi) $y = -3x + 1$

(vii) $x^2 + y^2 = a^2$

(viii) $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Limit: —

(*) (CT-1 ques):

$$\text{If } f(x) = \begin{cases} e^{\frac{-|x|}{2}} & ; 1 < x < 0 \\ x^2 & ; 0 \leq x < 2 \end{cases}$$

Find the limit at $x=0$.

continuity: —

(*) Define continuous function

(*) (CT-1 ques): If $f(x) = |x-1|$, then test the continuity at $x=1$.

Differentiability: —

(*) (CT-1 ques): If $f(x)$ is differentiable at $x=a$, then show that function is continuous at that point

(*) If (CT-1 ques): If $f(x) = \begin{cases} 1, & x < 0 \\ 1 + \sin x, & 0 \leq x < \frac{\pi}{2} \\ 2 + (x - \frac{\pi}{2})^2, & x \geq \frac{\pi}{2} \end{cases}$

check the differentiability of $f(x)$ at $x = \frac{\pi}{2}$.

Gamma-Beta function:

(*) Define Gamma and Beta function

(*) (CT-2 ques): show that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$.

(*) show that $\int_0^{\pi/2} \sin^p \theta \cdot \cos^q \theta \, d\theta = \frac{\Gamma\left(\frac{p+1}{2}\right) \Gamma\left(\frac{q+1}{2}\right)}{2 \Gamma\left(\frac{p+q+2}{2}\right)}$

(*) (CT-2 ques): using Gamma and Beta function evaluate $\int_0^{\pi/2} \sin^5 x \cdot \cos^4 x \, dx$

Maximum-Minimum:

(*) Find the critical point of $f(x) = 4x^3 - 9x^2 + 6x$

(*) Find the inflection point of $f(x) = \frac{x^3}{3} + ax^2 - 3ax$

(*) Find the maximum and minimum value of $f(x) = \frac{4}{x} + \frac{36}{2-x}$.

(*) (CT-2 ques): Find the maximum and minimum value of $f(x) = 2x^3 - 3x^2 - 12x$.

Rolle's thm and Mean-value thm:

(*) ~~State~~ State Rolle's theorem.

(*) verify Rolle's theorem for $f(x) = x^2 - 3x + 2$ in the interval $(1, 2)$.

(*) verify Rolle's theorem for $f(x) = (x-1)(x-2)(x-3)$ in the interval $(1, 3)$.

(*) State mean-value theorem.

(*) Find the value of c of the mean-value theorem $f(b) - f(a) = f'(c)(b-a)$ if $f(x) = (x-1)(x-2)(x-3)$ on interval $(0, 4)$.

▣ Indeterminate form (L'Hospital Rule):

(*) Define Indeterminate form

(*) State L'Hospital Rule

(*) (CT-2 ques): using L'Hospital Rule evaluate $\lim_{x \rightarrow 0} \left[\frac{1}{x} - \frac{1}{x} \log(1+x) \right]$.

▣ Area: —

(*) Find the area of a region bounded by parabola $y = 4x$ and straight line $y = 2x$.

(*) Find the area included between the cycloid $x = a(\theta + \sin \theta)$, $y = a(1 + \cos \theta)$ and its base.

(*) Find the area included between the cycloid $x = a(\theta - \sin \theta)$, $y = a(1 - \cos \theta)$ and its base.

Integration:

(*) Titas $\rightarrow$ (12.B): (41) $\int \frac{dx}{\sqrt{x-a}(b-x)}$

(43) $\int \frac{3x+2}{5x^2+2x+3} dx$

(*) Titas $\rightarrow$ (12.C): (53) $\int \frac{x^3}{\sqrt{x-1}}$

(57) $\int \frac{x dx}{(x+1)\sqrt{1+2x-x^2}}$

Increasing-Decreasing function and tangent-normal:

(*) Define increasing and decreasing function.

(*) Show that $f(x) = x^3 - 3x^2 + 3x$ increases at point $x=2$.

(*) Show that $f(x) = x^4 - 8x^2$ decreases at $x=1$.

(*) Find the eqn of tangent line and normal line of $f = x^2 + 1$ at point $(2,5)$.

▣ State and proof Fundamental theorem of calculus.

Or

State and proof Fundamental theorem of Integral calculus.